

Wahab Sohail

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PERSONAL PROFILE

Computer Engineering student with strong foundations in C++, software development, and real-time systems. Experienced in developing AI-powered and computer vision applications on Linux-based embedded platforms, with hands-on work in hardware integration, performance optimization, and real-time processing. Skilled in problem-solving, debugging, and building scalable software solutions for intelligent systems.

EDUCATION

National University of Science and Technology (NUST) 2022 – 2026
Bachelor's in Computer Engineering

WORK EXPERIENCE

Machine Learning Intern | RISETech Pvt. Ltd. July 2025 – Aug 2025

- Worked on **AI and data-driven research projects** involving data preprocessing, model training, and performance evaluation using ML and deep learning frameworks.
- Supported development of **intelligent healthcare and biomedical solutions**; gained hands-on experience with supervised/unsupervised learning, neural networks, and data visualization.

Intern | National Centre of Robotics and Automation (NCRA) Aug 2024 – Sept 2024

- Assisted in setting up and configuring development environments on Linux-based systems.
- Deployed applications on edge devices (**Raspberry Pi, Jetson Nano**) and optimized performance.
- Worked with GPU-based systems (**CUDA**) and managed system resources for compute-heavy tasks.
- Supported data handling, storage management, and experiment tracking.
- Collaborated in testing, debugging, and improving system reliability across different hardware setups.

FINAL YEAR PROJECT

Phoneme Level Speech Recognition with Error Analysis

Tech Stack: Python, C++, Raspberry Pi, Linux, Kaldi, Docker

- Deployed a real-time **Urdu phoneme recognizer** on Raspberry Pi 5 for speech rehabilitation – edge AI on constrained hardware.
- Designed acoustic modelling pipeline with **Kaldi**; optimized inference latency using **OpenBLAS** and hardware-specific build configs.
- Used a cross-language learning approach (English and Persian) to overcome the lack of Urdu data, improving phoneme recognition accuracy.
- Built a cross-platform mobile UI (**Kivy**) with live visual feedback and progress tracking for end-users.

PROJECTS

Tool Insights Chat

Tech: Next.js, TypeScript, Convex, Vercel AI SDK, Composio, Jira

- Built an AI chatbot integrating tools like **Gmail, Slack, and Jira** to deliver real-time insights.
- Applied Agile workflows using Jira with sprint-based task management and collaboration.
- Developed a scalable full-stack system with real-time data handling and streaming responses.

Sign Language Recognition System | Real-Time Hand Gesture Detection & Translation

Tech: Python, OpenCV, MediaPipe, CNN, LSTM, Streamlit

- Built a real-time system for detecting and translating hand gestures using webcam input.
- Combined **CNN** (for alphabets A-Z) and **LSTM** (for dynamic gestures) with **MediaPipe** hand tracking (21 keypoints).
- Applied smoothing techniques to improve prediction stability during live inference.

- Developed an interactive **Streamlit** app with text-to-speech output for real-time usability.

Santander Customer Transaction Prediction

Tech: Python, LightGBM, Naive Bayes, PyTorch, SHAP, Scikit-learn

- Built and compared multiple models (Residual MLP, LightGBM, Naive Bayes) on large tabular data.
- Selected Naive Bayes based on data analysis, focusing on efficiency and interpretability.
- Used **SHAP** to analyze feature importance and identify key predictors.
- Implemented a complete ML pipeline with preprocessing, cross-validation, and reproducibility.

Autonomous Navigation Using Image Processing

Tech: Python, OpenCV

- Built a self-driving car prototype using image processing for real-time navigation.
- Implemented lane detection using **edge detection, colour masking, and perspective transformation**.
- Enabled basic autonomous path following using vision-based decision-making.

Blood Management System

Tech: Node.js, MySQL

- Developed a database-driven system to manage donor and recipient records.
- Designed relational database schemas and implemented SQL queries.
- Built backend logic and handled CRUD operations with proper data validation.

TECHNICAL SKILLS

- **Programming & Scripting:** Python, C++, Java, SQL, Verilog, MATLAB
- **Operating Systems:** Linux (Ubuntu, Raspberry Pi OS), Windows
- **Networking & Systems:** Basic networking, router configuration, system deployment
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Tools & Platforms:** Git, GitHub, AWS S3, Cisco Packet Tracer, VS Code, Jira
- **Edge Computing Devices:** Jetson Nano, Raspberry Pi, STM32
- **AI & Machine Learning:** LLMs, Fine-Tuning, RAG, TensorFlow, PyTorch, Deep Learning, Computer Vision

LEADERSHIP & ACTIVITIES

President | COMPPEC, NUST

2025 – 2026

- Led end-to-end planning and execution of a large-scale computing event.
- Managed cross-functional teams and coordinated with sponsors, universities, and stakeholders.
- Oversaw budgeting, logistics, and promotional campaigns to ensure successful delivery.

Fund Manager | NUST Volunteer Club (NVC)

2025 – 2026

- Supervised transparent allocation and management of club funds to support charitable initiatives and individuals in need.

Volunteer | P@SHA

2024

- Coordinated end-to-end event operations from the evaluation phase to the closing ceremony, ensuring seamless workflow and effective team collaboration.

Volunteer | Ireland Expo 2024

2024

- Collaborated with event teams to organize and execute logistics, ensuring successful event delivery and stakeholder satisfaction.